

Magnetic Coupling in Nanographenes

Visual summary — 7-AGNR end states & the triangulene–phenalenyl spin chain

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7-AGNR

Motivation: topological end states in finite 7-AGNRs

7-AGNRs belong to the $N = 3p + 1$ family ($p = 2$): semiconducting, synthesized by on-surface polymerization/cyclodehydrogenation of DBBA precursors on Au(111).

Finite-length ribbons host **spin-polarized topological end states** (SPTs) at their zigzag-terminated ends, each carrying $S \approx 1/2$.

Origin: a nontrivial $\mathbb{Z}_2$ (Zak-phase) invariant of the bulk bands $\Rightarrow$ bulk–boundary correspondence guarantees in-gap boundary modes, independent of edge details.

Key question: what is the exchange coupling $J(L)$ between the two end spins, and how does it scale with ribbon length L (number of DBBA monomers)?

Third-nearest-neighbour tight-binding on the p_z lattice:

$$H_{\text{TB}} = - \sum_{\langle i,j \rangle} t_1 c_i^\dagger c_j - \sum_{\langle\langle i,j \rangle\rangle} t_2 c_i^\dagger c_j - \sum_{\langle\langle\langle i,j \rangle\rangle\rangle} t_3 c_i^\dagger c_j + \text{H.c.}$$

t_1	t_2	t_3	U (Hubbard)
-2.7 eV	-0.2 eV	-0.18 eV	3.0 eV

Mean-Field Hubbard (MFH): $H = H_{\text{TB}} + U \sum_i n_{i\uparrow} n_{i\downarrow}$, decoupled self-consistently (SCF tolerance 10^{-13} on the density matrix).

$$m_i = \langle n_{i\uparrow} \rangle - \langle n_{i\downarrow} \rangle \quad (\text{local magnetic moment})$$

Two independent routes to the exchange coupling J

1. Self-consistent MFH total energies

$$J = E_{\text{FM}} - E_{\text{AFM}}$$

AFM: open-shell singlet, $S_z = 0$. FM: triplet enforced via $q = (N/2+1, N/2-1)$, $S_z = 1$.

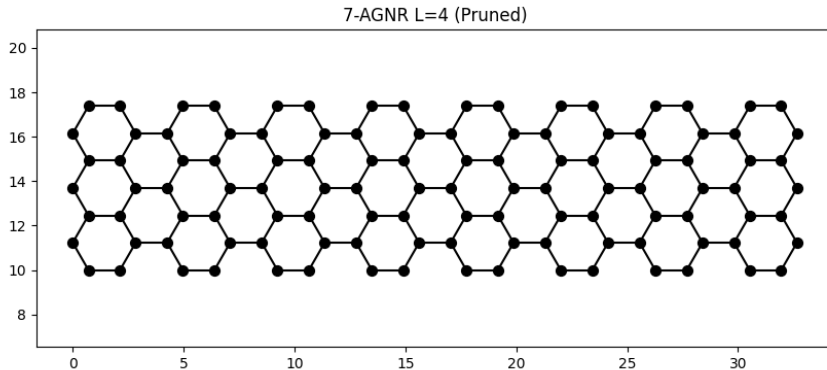
2. Effective Hubbard dimer (Anderson superexchange)

$$t_{\text{eff}} = \frac{1}{2}|\varepsilon_{\text{LUMO}} - \varepsilon_{\text{HOMO}}|, \quad U_{\text{eff}} = U \sum_i |\psi_L(i)|^4, \quad J = \frac{4 t_{\text{eff}}^2}{U_{\text{eff}}}$$

$\psi_L = (\psi_{\text{HOMO}} + \psi_{\text{LUMO}})/\sqrt{2}$: reconstructed localized end-state orbital.

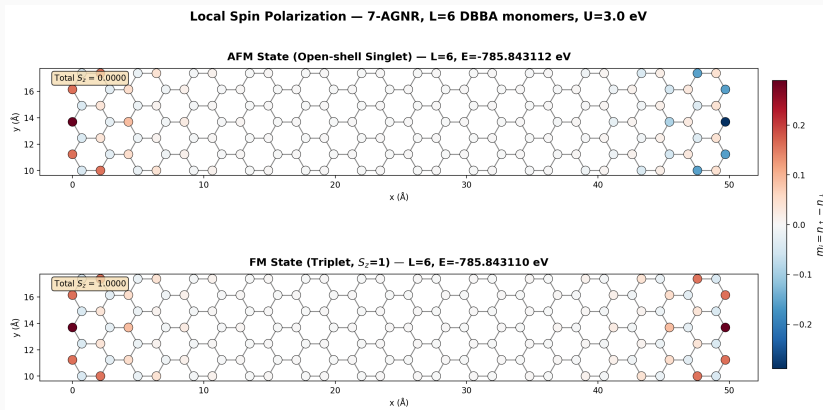
Since $t_{\text{eff}} \propto e^{-L/\xi_0}$: $J(L) = Ae^{-L/\xi}$ is predicted to decay exponentially.

Geometry check



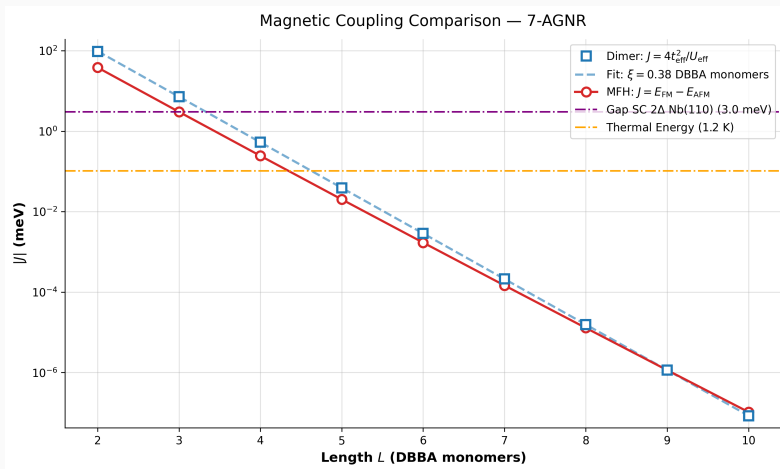
Pruned, zigzag-terminated 7-AGNR, $L = 4$ DBBA monomers.

Spin-polarized end states ($L = 6$)



AFM (top, $S_z=0$) vs. FM (bottom, $S_z=1$) — moments localized at the two ends.

Dimer vs. Heisenberg (MFH)



Both routes agree: $J(L) = Ae^{-L/\xi}$, $\xi = 0.38$ DBBA monomers.

Triangulene–Phenalenyl Copolymer

Motivation: chemically designed spin chains

Complementary strategy to ribbon end states: assemble a spin chain from **chemically localized radical building blocks**.

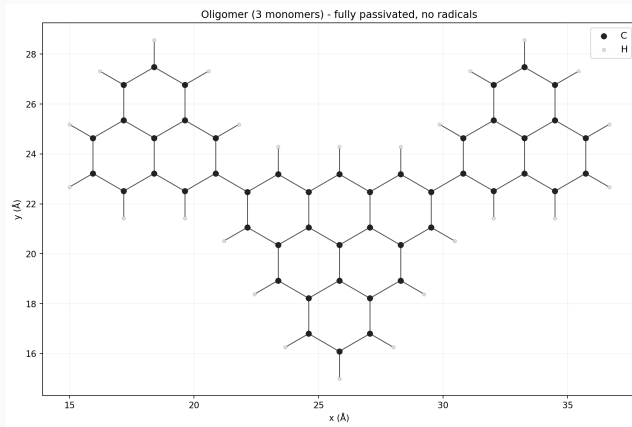
Repeating motif $P-T-P$ under open boundary conditions: $P-T-P-[P-T-P]_{N-1}$.

Phenalenyl (P): alternant radical, sublattice imbalance 1 $\Rightarrow$ single $S = \frac{1}{2}$ moment.

[3]Triangulene (T): sublattice imbalance 2 $\Rightarrow S = 1$ moment; represented as two $S = \frac{1}{2}$ sites T_a, T_b locked into a triplet by a strong *ferromagnetic* intra-unit exchange $J_T < 0$.

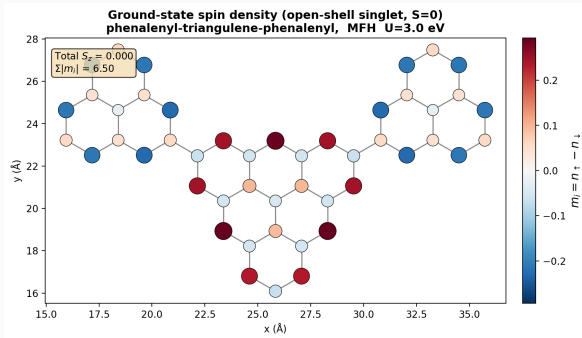
Every monomer maps onto four $S = \frac{1}{2}$ nodes $[P, T_a, T_b, P]$; the oligomer maps onto a chain of $n = 4N$ spin- $\frac{1}{2}$ sites.

Copolymer geometry



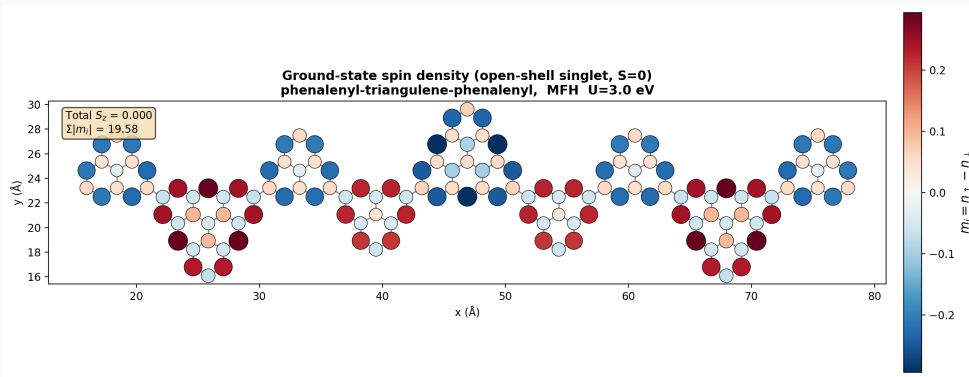
P-T-P oligomer, $N = 1$, closed-shell reference structure.

MFH ground-state spin density: single monomer ($N = 1$)



Open-shell singlet ground state, $S_z = 0$ (MFH, $U = 3.0$ eV): total $\sum_i |m_i| = 6.50$.
Spin density directly visualizes the P ($S=1/2$) and T ($S=1$, on T_a, T_b) building blocks predicted by sublattice-imbalance counting.

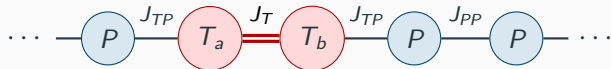
MFH spin density (trimer, $N = 3$)



Open-shell singlet, $S_z = 0$; alternating P/T moments along the chain.

From MFH to an effective Heisenberg spin chain

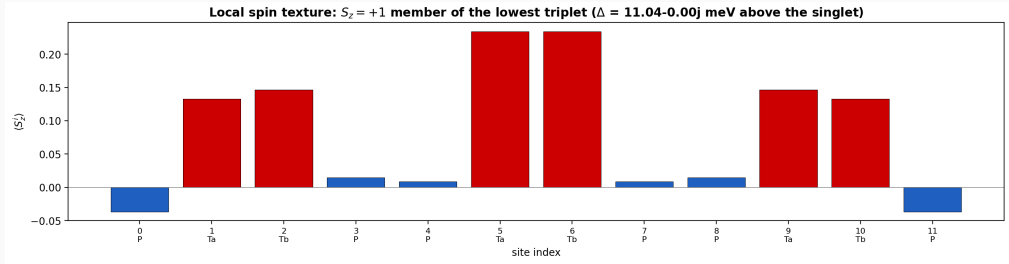
$$H = \sum_{m=0}^{N-1} \left[J_{TP} (\mathbf{S}_{4m} \cdot \mathbf{S}_{4m+1} + \mathbf{S}_{4m+2} \cdot \mathbf{S}_{4m+3}) + J_T \mathbf{S}_{4m+1} \cdot \mathbf{S}_{4m+2} \right] + J_{PP} \sum_{m=0}^{N-2} \mathbf{S}_{4m+3} \cdot \mathbf{S}_{4m+4}$$



J_{PP} (AFM)	J_{TP} (AFM)	J_T (FM)
38 meV	40 meV	-500 meV

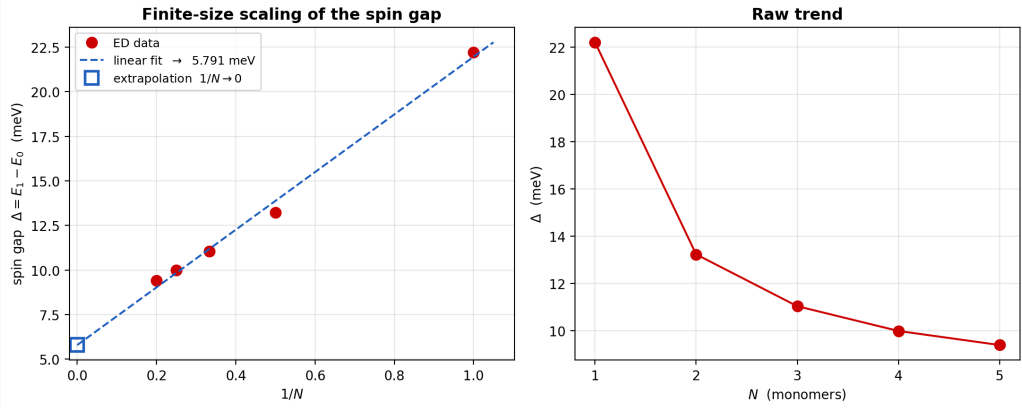
Since $|J_T| \gg J_{TP}, J_{PP}$: each (T_a, T_b) pair freezes into an effective $S = 1$ object $\Rightarrow$
 mixed $\frac{1}{2}-1-\frac{1}{2}$ AFM spin chain.

Exact-diagonalization spin texture



Singlet ground state; lowest triplet shown — singlet–triplet gap $\Delta = 11.04$ meV.

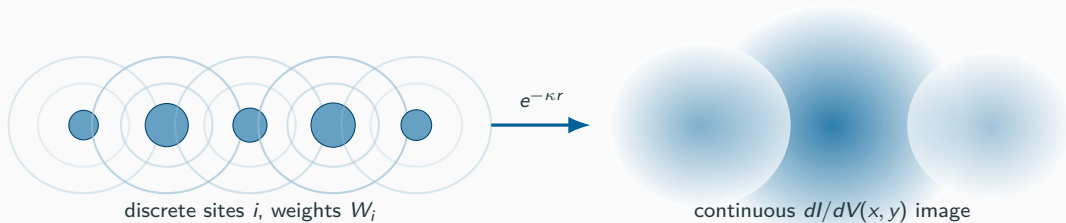
Finite-size scaling: a robust spin gap



ED sweep $N = 1-5$: Δ vs. $1/N$ (extrapolation) and Δ vs. N (raw trend).

$\Delta(1/N \rightarrow 0) = 5.791$ meV: a robust, non-zero gap in the thermodynamic limit — a gapped, non-trivial collective magnetic phase.

From spin nodes to STM images: the Tersoff–Hamann bridge

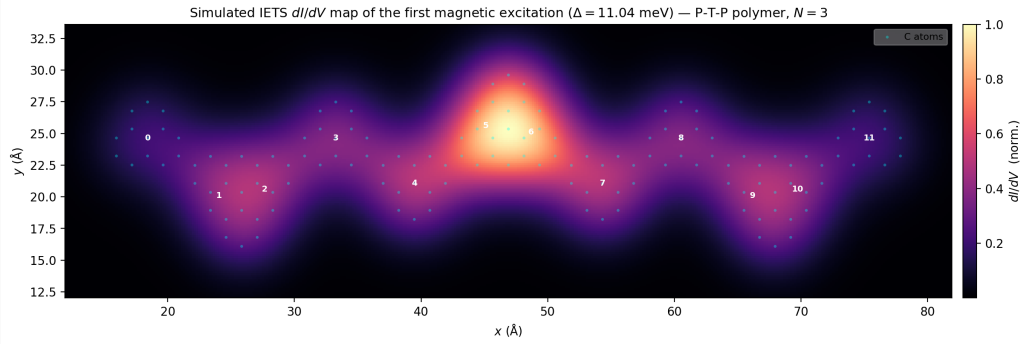


$$\frac{dI}{dV}(x, y) \propto \sum_i W_i \sum_{\alpha \in \text{site } i} \exp(-\kappa r_\alpha(x, y))$$

$W_i = \sum_m \sum_{\mu=x,y,z} |\langle 1_m | S_i^\mu | 0 \rangle|^2$: spin-flip spectral weight of each site, spread over its **real carbon coordinates**.

Exponential decay κ (constant-height vacuum tunneling) acts as a spatial broadening kernel.

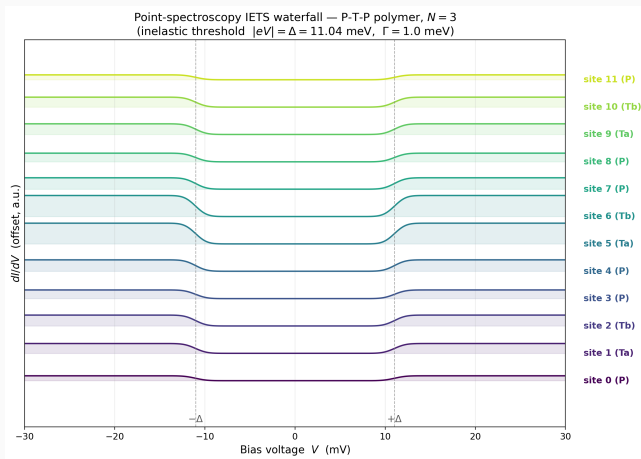
Simulated spatial IETS map ($N = 3$, $\Delta = 11.04$ meV)



Lighthouse effect: excitation localized on the central [3]triangulene core (sites 5, 6).

Dark interference regions on the internal phenalenyl linkers — signature of spin frustration.

Point-spectroscopy IETS waterfall: where to park the tip



Sites 5, 6 maximal; sites 3, 4, 7, 8 quenched — a roadmap for tip placement.

To be continued

